

02477 – Bayesian Machine Learning: Lecture 2

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Outline

- 1 Quick re-cap of last week
- 2 Probabilistic machine learning
- 3 The plug-in approximation
- 4 Grid approximations for non-conjugate models
- 5 Introduction to exercise: towards logistic regression

Quick re-cap of last week

Quick re-cap of Beta-binomial model and what's next

- *Bayes' rule* provides a systematic way to combine data with prior knowledge

$$p(\theta|y) = \frac{p(y|\theta)p(\theta)}{p(y)}$$

- *The beta-binomial model* is a *conjugate* model

$$p(\theta) = \text{Beta}(\theta|a_0, b_0) \quad (\text{Prior})$$

$$p(y|\theta) = \binom{N}{y} \theta^y (1 - \theta)^{N-y} \quad (\text{Likelihood})$$

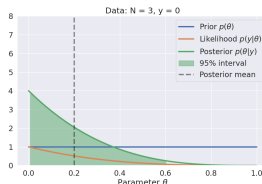
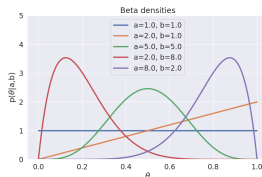
$$p(\theta|y) = \text{Beta}(\theta|y + a_0, N - y + b_0) \quad (\text{Posterior})$$

- Estimate θ using the *mean* of the *posterior distribution*

$$\theta_{\text{Bayes}} = \mathbb{E}[\theta|y] \equiv \int \theta p(\theta|y) d\theta$$

- .. and use *credibility intervals* of the posterior to quantify the uncertainty

$$P(\theta \in [0.01, 0.60] | y) = 0.95$$



What about making predictions?

Example continued: suppose we have this website ad with $N = 3$ views and $y = 0$ clicks.

- Using a *uniform prior*, i.e. $p(\theta) = \text{Beta}(\theta|1, 1)$

$$p(\theta) = \text{Beta}(\theta|1, 1) \quad (\text{Prior})$$

$$p(y|\theta) = \binom{3}{0} \theta^0 (1 - \theta)^3 \quad (\text{Likelihood})$$

$$p(\theta|y) = \text{Beta}(\theta|1, 4) \quad (\text{Posterior})$$

- *Summarize* our knowledge using posterior

$$\mathbb{E}[\theta|y] = \frac{1}{5}, \quad P(\theta \in [0.01, 0.60] | y) \approx 0.95$$

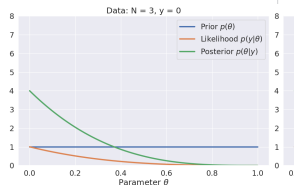
- **Goal:** *predict* number of clicks y^* in the next $N^* = 50$ views?

$$p(y^*|N^*, \theta) = \text{Bin}(y^*|N^*, \theta)$$

- Recall: mean of a Binomial random variable with prob. θ :

$$\mathbb{E}[y^*] = N^* \theta = 50\theta$$

Which value of θ to use? How do we use the *posterior knowledge* to make predictions?
How do we take the *uncertainty* into account?



Probabilistic machine learning

Probabilistic machine learning I

Product rule

$$p(\mathbf{a}, \mathbf{b}) = p(\mathbf{b}|\mathbf{a})p(\mathbf{a})$$

Sum rule

$$p(\mathbf{b}) = \int p(\mathbf{a}, \mathbf{b})d\mathbf{a}$$

Conditional

$$p(\mathbf{a}|\mathbf{b}) = \frac{p(\mathbf{a}, \mathbf{b})}{p(\mathbf{b})}$$

Conditional independence

$$p(\mathbf{a}, \mathbf{b}|\mathbf{c}) = p(\mathbf{a}|\mathbf{c})p(\mathbf{b}|\mathbf{c})$$

- A probabilistic model is *completely specified* by its *joint distribution*
- Consider a model with two *random variables*: y (data) and θ (unknown parameter)
- The *joint distribution* of all random variables can be expressed via the *product rule*

$$p(\theta, y) = p(y|\theta)p(\theta)$$

- The *posterior distribution* can be obtained by *conditioning* on y

$$p(\theta|y) = \frac{p(y, \theta)}{p(y)} = \frac{p(y|\theta)p(\theta)}{p(y)}$$

- The *evidence* $p(y)$ can be obtained from the joint distribution via the *sum rule*

$$p(y) = \int p(y, \theta)d\theta = \int p(y|\theta)p(\theta)d\theta$$

- Hence, in theory, we can derive *all quantities of interest* from the *joint distribution*

Probabilistic machine learning II

Product rule

$$p(\mathbf{a}, \mathbf{b}) = p(\mathbf{b}|\mathbf{a})p(\mathbf{a})$$

Sum rule

$$p(\mathbf{b}) = \int p(\mathbf{a}, \mathbf{b})d\mathbf{a}$$

Conditional

$$p(\mathbf{a}|\mathbf{b}) = \frac{p(\mathbf{a}, \mathbf{b})}{p(\mathbf{b})}$$

Conditional independence

$$p(\mathbf{a}, \mathbf{b}|\mathbf{c}) = p(\mathbf{a}|\mathbf{c})p(\mathbf{b}|\mathbf{c})$$

- A probabilistic model is *completely specified* by its *joint distribution*

$$p(\theta, y) = p(y|\theta)p(\theta)$$

- What if we have more than one observed variable, e.g. y_1 and y_2 ?
- For a *broad class of models* the likelihood can be further decomposed using *conditional independence*:

$$p(y_1, y_2|\theta) = p(y_1|\theta)p(y_2|\theta)$$

- ... or more generally for $\mathbf{y} = [y_1 \quad y_2 \quad \dots \quad y_N]$

$$p(\mathbf{y}|\theta) = p(y_1|\theta)p(y_2|\theta) \dots p(y_N|\theta) = \prod_{n=1}^N p(y_n|\theta)$$

- The *joint distribution* becomes

$$p(\theta, \mathbf{y}) = p(\mathbf{y}|\theta)p(\theta) = \prod_{n=1}^N p(y_n|\theta)p(\theta)$$

Website ad example continued: Making predictions

- **Example continued:** Your website ad has been shown $N = 123$ times and generated $y = 12$ clicks. Suppose you pay for another $N^* = 50$ views, how many clicks y^* should you expect *given the observed data*?
- Assuming each click can be modelled using *conditionally independent* Bernoulli trials with the *same probability* θ

$$p(y|\theta) = \text{Bin}(y|N, \theta)$$

$$p(y^*|\theta) = \text{Bin}(y^*|N^*, \theta)$$

- The assumption of *conditional independence* implies

$$p(y, y^*|\theta) = p(y|\theta)p(y^*|\theta) = \text{Bin}(y|N, \theta)\text{Bin}(y^*|N^*, \theta)$$

- Completing the model by *imposing a prior* for θ

$$p(\theta) = \text{Beta}(\theta|a_0, b_0)$$

- **Goal:** compute *predictive distribution* of y^* given we have observed $y = 12$, i.e. $p(y^*|y = 12)$.

A probabilistic perspective on making predictions

Product rule

$$p(\mathbf{a}, \mathbf{b}) = p(\mathbf{b}|\mathbf{a})p(\mathbf{a})$$

Sum rule

$$p(\mathbf{b}) = \int p(\mathbf{a}, \mathbf{b})d\mathbf{a}$$

Conditional

$$p(\mathbf{a}|\mathbf{b}) = \frac{p(\mathbf{a}, \mathbf{b})}{p(\mathbf{b})}$$

Conditional independence

$$p(\mathbf{a}, \mathbf{b}|\mathbf{c}) = p(\mathbf{a}|\mathbf{c})p(\mathbf{b}|\mathbf{c})$$

Goal: Given some data y , what can we say about a new observation y^* ?

- Step 1: Formulate *joint distribution* for *all variables* of interests

$$p(y^*, y, \theta) = p(y^*, y|\theta)p(\theta) = p(y^*|\theta)p(y|\theta)p(\theta)$$

- Step 2: *Condition* on the *observed data* y

$$p(y^*, \theta|y) = \frac{p(y^*, y, \theta)}{p(y)} = \frac{p(y^*|\theta)p(y|\theta)p(\theta)}{p(y)}$$

- Step 3: *Marginalize* out parameter θ using the *sum rule* to get the *posterior predictive distribution*

$$p(y^*|y) = \int p(y^*, \theta|y)d\theta = \int \frac{p(y^*|\theta)p(y|\theta)p(\theta)}{p(y)}d\theta = \int p(y^*|\theta)p(\theta|y)d\theta = \mathbb{E}_{p(\theta|y)} [p(y^*|\theta)]$$

- **Key take-away:** To reason about y^* *given* y , we need to *average the likelihood* for y^* wrt. to the *posterior distribution* $p(\theta|y)$.

Quiz time

Take the quiz called
Lecture 2: Prior, likelihood, posterior, posterior predictive
to test your understanding.

Website example

- **Example continued:** Your website ad has been shown $N = 123$ times and generated $y = 12$ clicks. Suppose you pay for another $N^* = 50$ views, how many clicks y^* should you expect *given the observed data*?

- We already defined the model

$$p(y|\theta) = \text{Bin}(y|N, \theta) \quad (\text{Likelihood})$$

$$p(y^*|\theta) = \text{Bin}(y^*|N^*, \theta) \quad (\text{Predictive likelihood})$$

$$p(\theta) = \text{Beta}(\theta|a_0, b_0) \quad (\text{Prior})$$

- We know how to compute the *posterior distribution*

$$p(\theta|y) = \text{Beta}(\theta|y + a_0, N - y + b_0)$$

- Next, we want to compute the *posterior predictive distribution*

$$p(y^*|y) = \int p(y^*|\theta)p(\theta|y)d\theta = \int \text{Bin}(y^*|N^*, \theta)\text{Beta}(\theta|y + a_0, N - y + b_0)d\theta$$

- **Intuition:** Instead of plugging in a single value for the parameter estimate, we plug in all possible values for θ and weight the result according to $p(\theta|y)$

Website example

- Compute *posterior predictive distribution*

$$\begin{aligned} p(y^* = k|y) &= \int \text{Bin}(y = k|N^*, \theta) \text{Beta}(\theta|y + a_0, N - y + b_0) d\theta \\ &= \int \binom{N^*}{k} \theta^k (1 - \theta)^{N^* - k} \frac{1}{B(\alpha, \beta)} \theta^{\alpha - 1} (1 - \theta)^{\beta - 1} d\theta \\ &= \binom{N^*}{k} \frac{1}{B(\alpha, \beta)} \int \theta^k (1 - \theta)^{N^* - k} \theta^{\alpha - 1} (1 - \theta)^{\beta - 1} d\theta && \text{(Linearity)} \\ &= \binom{N^*}{k} \frac{1}{B(\alpha, \beta)} \int \theta^{k + \alpha - 1} (1 - \theta)^{\beta + N^* - k - 1} d\theta && \text{(Simplify)} \\ &= \binom{N^*}{k} \frac{1}{B(\alpha, \beta)} \int \theta^{k + \alpha - 1} (1 - \theta)^{\beta + N^* - k - 1} d\theta \end{aligned}$$

- We recognize the terms in green as the *functional form* of a Beta density, and hence, we know how to compute the integral

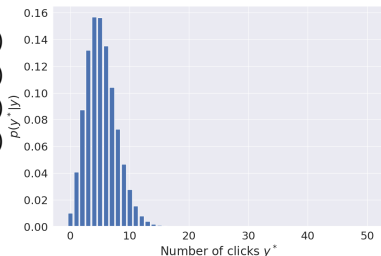
$$p(y^* = k|y) = \binom{N^*}{k} \frac{B(\alpha + k, \beta + N^* - k)}{B(\alpha, \beta)}$$

for $\alpha = y + a_0$ and $\beta = N - y + b_0$.

Website example: putting everything together

Example continued: Your website ad has been shown $N = 123$ times and generated $y = 12$ clicks. Suppose you pay for another $N^* = 50$ views, how many clicks y^* should you expect *given the observed data*?

$$\begin{aligned} p(\theta) &= \text{Beta}(\theta|1, 1) && \text{(Prior)} \\ p(y|\theta) &= \text{Bin}(y|123, \theta) && \text{(Likelihood)} \\ p(y^*|\theta) &= \text{Bin}(y^*|50, \theta) && \text{(Predictive likelihood)} \\ p(\theta|y) &= \text{Beta}(\theta|13, 112) && \text{(Posterior)} \end{aligned}$$



- Distribution of clicks y^* based on views $N^* = 50$ views

$$p(y^* = k|y) = \binom{N^*}{k} \frac{B(\alpha + k, \beta + N^* - k)}{B(\alpha, \beta)} = \binom{50}{k} \frac{B(13 + k, 162 - k)}{B(13, 112)}$$

- The expected number of clicks given the data is

$$\mathbb{E}_{p(y^*|y)}[y^*] = \sum_{k=0}^{50} k p(y^* = k|y) \approx 5.2$$

The plug-in approximation

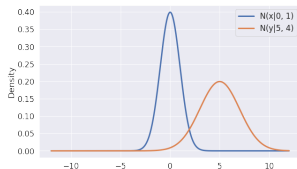
A few prerequisites first

Univariate Gaussians

- The *normal distribution* (also known as the Gaussian) is distribution over $x \in \mathbb{R}$ with density

$$\mathcal{N}(x|\mu, \sigma^2) = \frac{1}{\sqrt{2\pi\sigma^2}} \exp\left(-\frac{(x - \mu)^2}{2\sigma^2}\right)$$

- Two parameters: $\mu \equiv \mathbb{E}[x]$ and $\sigma^2 \equiv \mathbb{V}[x]$
- Widely popular due to Central limit theorems, maximum entropy principle, relation to least squares minimization, nice mathematical properties
- We will talk more about Gaussians later in this course



A few prerequisites first

Dirac's delta function

- Consider a Gaussian random variable $x \sim \mathcal{N}(\mu, \sigma^2)$.
In the *limit* $\sigma^2 \rightarrow 0$, x is effectively a *constant* $x = \mu$:
- We say that x follows a *Dirac's delta distribution* centered at μ

$$p(x) = \lim_{\sigma^2 \rightarrow 0} \mathcal{N}(x|\mu, \sigma^2) = \delta(x - \mu)$$

- Important properties

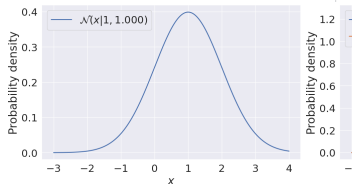
$$\delta(x - \mu) = \begin{cases} \infty & \text{if } x = \mu \\ 0 & \text{otherwise} \end{cases} \quad (1)$$

$$\int \delta(x - \mu) dx = 1 \quad (2)$$

$$\int f(x) \delta(x - \mu) dx = f(\mu) \quad (3)$$

- Eq. (3) is called the *sifting property* and implies

$$\mathbb{E}_{\delta(x-\mu)} [f(x)] = f(\mu)$$



The plugin approximation

- We showed that the rules of probability theory dictates that

$$p(y^*|y) = \int p(y^*|\theta)p(\theta|y)d\theta$$

- While this is *optimal* given the model, it can be *non-trivial* in practice
- If we *assume* that there is a *single best parameter* $\hat{\theta}$, e.g. $\hat{\theta}_{\text{MLE}}$ or $\hat{\theta}_{\text{MAP}}$, then we can approximate $p(\theta|y)$ using a *Dirac's delta function* $\delta(\theta - \hat{\theta})$

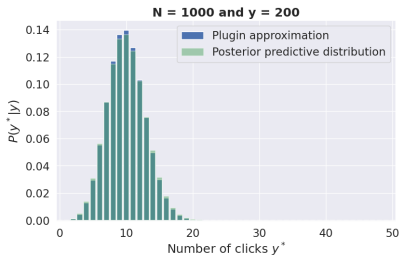
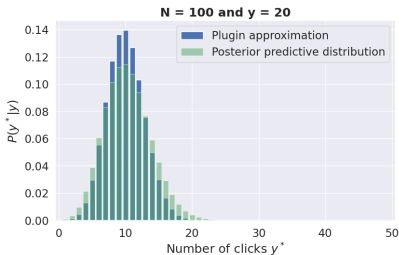
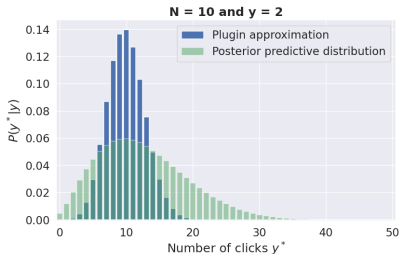
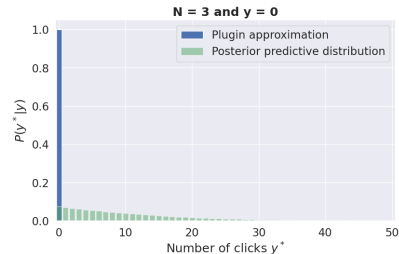
$$p(\theta|y) \approx \delta(\theta - \hat{\theta})$$

- Using the *sifting property* of Dirac's delta implies

$$p(y^*|y) = \int p(y^*|\theta)p(\theta|y)d\theta \approx \int p(y^*|\theta)\delta(\theta - \hat{\theta})d\theta = p(y^*|\hat{\theta})$$

- Therefore, this is called a *plug-in approximation*.
- Very *easy to compute*, but *ignores uncertainty* for our estimate of θ , and hence, often *producing overconfident predictions*.
- This is how we make predictions in deep learning...

The posterior predictive distribution and plugin approximations



Grid approximations for non-conjugate models

Introduction to non-conjugate models

Big picture so far...

- We studied the binomial model for estimating proportions and imposed a Beta prior for θ for Bayesian inference
- We derived the *posterior* and *posterior predictive* distributions *analytically*. This is possible due to *conjugacy* of the Beta prior and binomial likelihood
- Supposed our analysis required computing the posterior mean for a *different prior*, e.g. $p(\theta) = \frac{1}{Z} e^{\sin(\pi\theta^2)}$

$$\mathbb{E}[\theta|y] = \int \theta p(\theta|y) d\theta = \int \theta \frac{p(y|\theta)p(\theta)}{p(y)} d\theta$$

- To compute the posterior mean, variance etc. we need the evidence $p(y)$

$$\begin{aligned} p(y) &= \int p(y|\theta)p(\theta)d\theta = \int \text{Bin}(y|N, \theta) \frac{1}{Z} e^{\sin(\pi\theta^2)} d\theta \\ &= \int \binom{N}{y} \theta^y (1-\theta)^{N-y} \frac{1}{Z} e^{\sin(\pi\theta^2)} d\theta \\ &= \binom{N}{y} \frac{1}{Z} \int \theta^y (1-\theta)^{N-y} e^{\sin(\pi\theta^2)} d\theta = ? \end{aligned}$$

- Unfortunately, we *cannot* evaluate the evidence, i.e. $p(y)$ analytically *intractable* for most models of practical interest...

Approximate inference methods

- **In this course:** We will study several computational tools and approximation inference methods for dealing with such *intractable distributions*
 1. Grid approximations
 2. Laplace approximations
 3. Variational inference
 4. Markov Chain Monte Carlo methods
- **Goal for these methods:** Compute *tractable approximation* $q(\theta)$ of true posterior $p(\theta|y)$ such that
 1. $q(\theta)$ resembles the true posterior, i.e. $p(\theta|y) \approx q(\theta)$
 2. $q(\theta)$ should be tractable s.t. we can compute posterior summaries, predictions etc.
- **This week:** We will focus on *grid approximations*, which are easy to understand and apply, and they will help build our intuition about marginalization.

The grid approximation

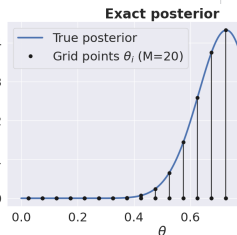
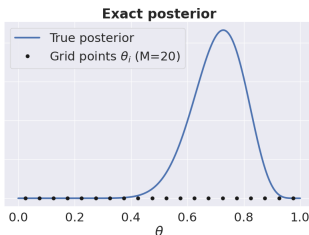
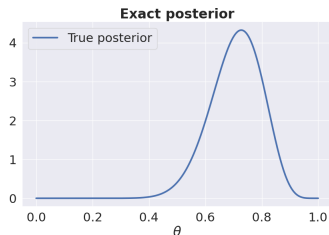
■ Constructing the grid approximation for $p(\theta|y)$

1. We define a set of grid points for θ : $0 \leq \theta_1 < \theta_2 < \dots < \theta_M \leq 1$
2. We evaluate the exact posterior (up to a constant) at all the grid points, i.e.

$$\tilde{\pi}_i \propto p(\theta_i|y) \propto p(y|\theta_i)p(\theta_i)$$

3. Sum all values to get normalization constant $Z = \sum_{i=1}^M \tilde{\pi}_i$
4. Compute normalized probabilities $\pi_i = \frac{1}{Z} \tilde{\pi}_i$ to get the grid approximation

$$q(\theta) = \sum_{i=1}^M \pi_i \delta(\theta - \theta_i)$$



The grid approximation

Posterior summaries

- The grid approximation is a discrete distribution, so computing summaries is easy, e.g the posterior mean

$$\mathbb{E}_{p(\theta|y)} [\theta] \approx \mathbb{E}_{q(\theta)} [\theta] = \sum_{i=1}^M \theta_i \pi_i$$

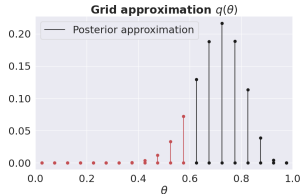
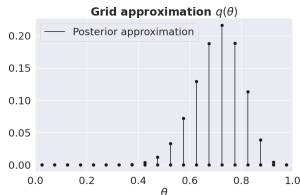
- ... and general expectations of $f(\theta)$

$$\mathbb{E}_{p(\theta|y)} [f(\theta)] \approx \mathbb{E}_{q(\theta)} [f(\theta)] = \sum_{i=1}^M f(\theta_i) \pi_i$$

- Example: Computing post. probabilities for $\theta < 0.6$

$$\begin{aligned} p(\theta < 0.6|y) &\approx q(\theta < 0.6) \\ &= \sum_{i=1}^M \mathbb{I}[\theta_i < 0.6] \pi_i \\ &= \sum_{i=1}^j \pi_i, \quad j = \max \{i | \theta_i < 0.6\} \end{aligned}$$

$$q(\theta) = \sum_{i=1}^M \pi_i \delta(\theta - \theta_i)$$



The grid approximation

The posterior predictive distribution

- General expectations of $f(\theta)$

$$\mathbb{E}_{p(\theta|y)} [f(\theta)] \approx \mathbb{E}_{q(\theta)} [f(\theta)] = \sum_{i=1}^M f(\theta_i) \pi_i$$

- The posterior predictive distribution $p(y^*|y)$

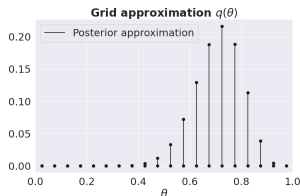
$$p(y^* = k|y) = \mathbb{E}_{p(\theta|y)} [\text{Bin}(y^* = k|N^*, \theta)]$$

- Hence, setting $f(\theta) = \text{Bin}(y^* = k|N^*, \theta_i)$ yields

$$p(y^* = k|y) \approx \sum_{i=1}^M \text{Bin}(y^* = k|N^*, \theta_i) \pi_i$$

- To make predictions we literally compute a weighted sum of all possible parameter values

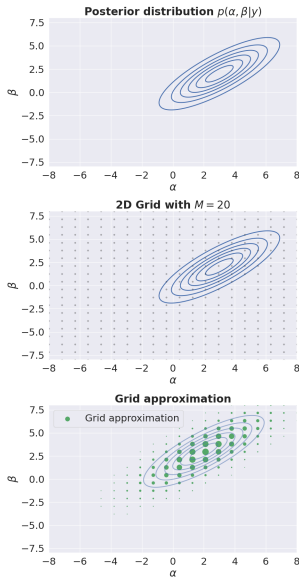
$$q(\theta) = \sum_{i=1}^M \pi_i \delta(\theta - \theta_i)$$



The grid approximation

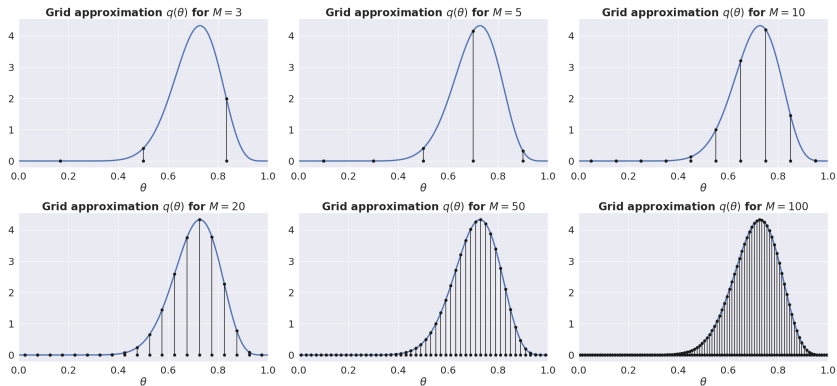
A few practical considerations

- Choosing the grid range
 - Range for $\theta \in [0, 1]$ is easy
 - For a different parameter $\alpha \in \mathbb{R}$, we need to choose an interval $[a, b]$ for the grid. Often identified visually.
- Scaling with model dimensionality
 - Suppose we use $M = 20$ points for each dimension
 - 1D: 20 evaluations
 - 2D: $20^2 = 400$ evaluations
 - 3D: $20^3 = 8000$ evaluations
 - 4D: $16000^4 = 160000$ evaluations
 - Grid approximations do not scale well beyond 3-4 dimensions
- Number of grid points M
 - M is balance between computational cost and accuracy
 - Grid approximation is zero when evaluated outside the grid points
 - Often diminishing returns as M increases (next slide)



The grid approximation

Summary



- **Pros:** Simple, easy and intuitive.
- **Cons:** Suffers from curse of dimensionality and does not scale beyond 3-4 dimensions

Introduction to exercise: towards logistic regression

Towards logistic regression

- So far we focussed on modelling *proportions*, i.e. $\theta \in [0, 1]$ given data about y successes in N *conditionally independent trials*
- The binomial likelihood is also often used in *dose-response* models, which is key for determining "safe" dosages for drugs, pollution, foods etc.
- **Example:** A company wants to study side effects of their new drug

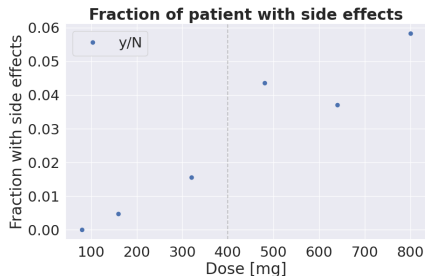
x (Dose in mg)	y (# side effects)	N (# patients)
80	0	69
160	4	832
320	13	835
480	20	459
640	12	324
800	6	103

- We could analyze the data for each dose independently using a beta-binomial model, but we would like to ...
 1. understand how dose affect the probability of side effects
 2. make predictions for new dosages x^*
 3. borrow "statistical strength" across dosages

Towards logistic regression

Example & motivation II

x (Dose in mg)	y (# side effects)	N (# patients)	y/N
80	0	69	0
160	4	832	0.005
320	13	835	0.016
480	20	459	0.044
640	12	324	0.037
800	6	103	0.058



- How accurate can we predict the probability of side effects for $x^* = 400\text{mg}$?

Towards logistic regression

Setting up the likelihood

- For each dose x_i , we assume

$$y_i|x_i \sim \text{Bin}(y_i|N_i, \theta_i), \quad \theta_i \in [0, 1]$$

- We model the probability θ_i as function of the dose x_i , i.e.

$$\theta_i \equiv \theta(x_i) = \sigma(\alpha + \beta x),$$

where $\sigma(x) : \mathbb{R} \rightarrow [0, 1]$ is a sigmoid function and $\alpha, \beta \in \mathbb{R}$ are model parameters.

- The likelihood of the i 'th observation (x_i, y_i)

$$p(y_i|x_i, \alpha, \beta) = \text{Bin}(y_i|N_i, \theta_i)$$

- Assuming conditional independence we can write the joint likelihood

$$p(\mathbf{y}|\mathbf{x}, \alpha, \beta) = \prod_{i=1}^M p(y_i|x_i, \alpha, \beta) = \prod_{i=1}^M \text{Bin}(y_i|N_i, \theta_i),$$

where $\mathbf{y} = [y_1, y_2, \dots, y_6]$ and similar for $\mathbf{x} = [x_1, x_2, \dots, x_6]$

- The predictive likelihood for y^* is

$$p(y^*|x^*, \alpha, \beta) = \text{Bin}(y^*|N_i, \theta^*)$$

where $\theta^* \equiv \theta(x^*)$

Towards logistic regression

Setting up the prior

- The model parameters are α (intercept) and β (slope) of the generalized linear model
- Prior information: we have no prior information about the sign of the parameters. Hence, we choose a zero-mean Gaussian distributions

$$p(\alpha, \beta) = \mathcal{N}(\alpha|0, \sigma_\alpha^2) \mathcal{N}(\beta|0, \sigma_\beta^2), \quad \sigma_\alpha^2, \sigma_\beta^2 > 0$$

- We can now write the *joint distribution* of $\alpha, \beta, \mathbf{y}, \mathbf{y}^*$ using the *product rule*

$$p(\mathbf{y}, \mathbf{y}^*, \alpha, \beta | \mathbf{x}, \mathbf{x}^*) = p(\mathbf{y}^* | \mathbf{x}^*, \alpha, \beta) p(\mathbf{y} | \mathbf{x}, \alpha, \beta) p(\alpha, \beta)$$

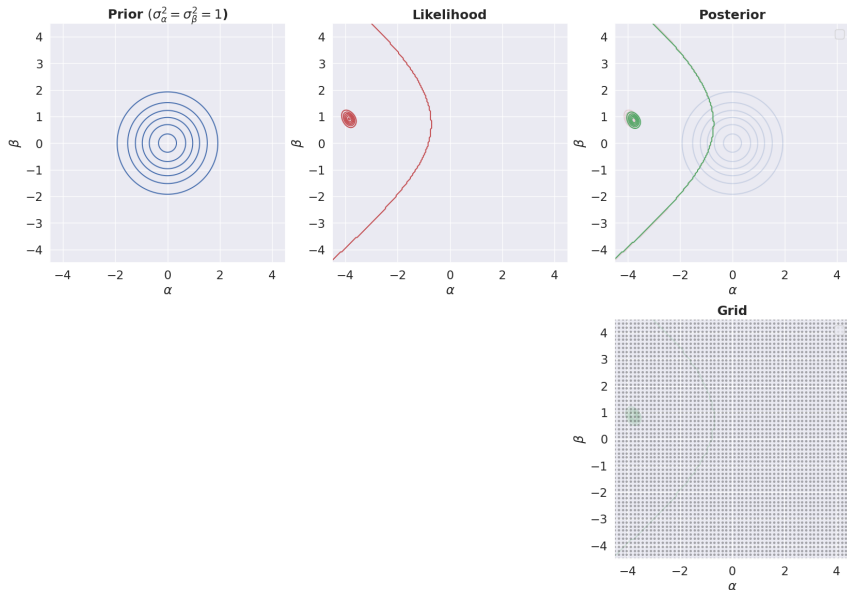
- The *posterior predictive distribution* is by *conditioning on $\mathbf{y}$* and *marginalizing out the parameters* via the sum rule

$$p(\mathbf{y}^* | \mathbf{y}, \mathbf{x}, \mathbf{x}^*) = \iint p(\mathbf{y}^*, \alpha, \beta | \mathbf{y}, \mathbf{x}, \mathbf{x}^*) d\alpha d\beta = \iint \underbrace{p(\mathbf{y}^* | \mathbf{x}^*, \alpha, \beta)}_{\text{likelihood for } \mathbf{y}^*} \underbrace{p(\alpha, \beta | \mathbf{y}, \mathbf{x})}_{\text{posterior distribution}} d\alpha d\beta$$

- After obtaining the posterior of α, β , we can *propagate* the posterior uncertainty of the parameters to any quantity that depends on α, β , i.e. $\theta(x) = \sigma(\alpha + \beta x)$, the fraction of people with side effects $\mathbf{y}^*/N$ etc.

Towards logistic regression

Visualizing the distributions

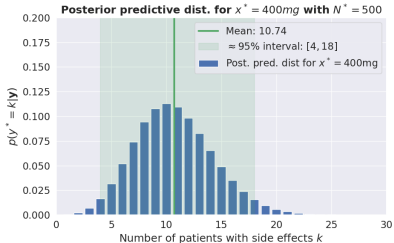


Towards logistic regression

Making predictions

Computing the posterior predictive distribution $p(y^* = k | \mathbf{y}, \mathbf{x}, x^*)$ for $x^* = 400\text{mg}$ and $N^* = 500$

$$\begin{aligned} p(y^* = k | \mathbf{y}, \mathbf{x}, x^*) &= \iint \underbrace{p(y^* = k | x^*, \alpha, \beta)}_{\text{likelihood for } y^*} \underbrace{p(\alpha, \beta | \mathbf{y}, \mathbf{x})}_{\text{posterior distribution}} d\alpha d\beta && \text{(Sum rule)} \\ &= \mathbb{E}_{p(\alpha, \beta | \mathbf{y}, \mathbf{x}, x^*)} [p(y^* = k | x^*, \alpha, \beta)] && \text{(Integrals as expectation)} \\ &= \mathbb{E}_{p(\alpha, \beta | \mathbf{y}, \mathbf{x}, x^*)} [\text{Bin}(y^* | N^*, \theta^*)] && \text{(Inserting dist.)} \\ &\approx \mathbb{E}_{q(\alpha, \beta)} [\text{Bin}(y^* | N^*, \theta^*)] && \text{(Grid approx.)} \\ &= \sum_{i,j} \text{Bin}(y^* | N^*, \sigma(\alpha_i + \beta_j x^*)) \pi_{ij} \end{aligned}$$



Intro to exercise

- On DTU Learn you will find an exercise for each week in notebook format
- We will spend all 4 hours from 13-17 working with the exercises
- In this exercise you will
 - Dive deeper into the Bayesian framework
 - Study and implement the probabilistic model for logistic regression for the Challenger Distaster dataset
 - Study and implement the grid approximations
 - Practice probabilistic reasoning
- Mix of pen&paper, programming and discussion questions
- Feel free to collaborate with your peers
- Ask for help!
 - Ask for help when stuck
 - Use teachers/TAs to check your understanding
 - Engage in discussion to practice
- Feedbacks persons: Meet at 16:45